

Week 7 Report: Model Development (end) and Model Evaluation (start)

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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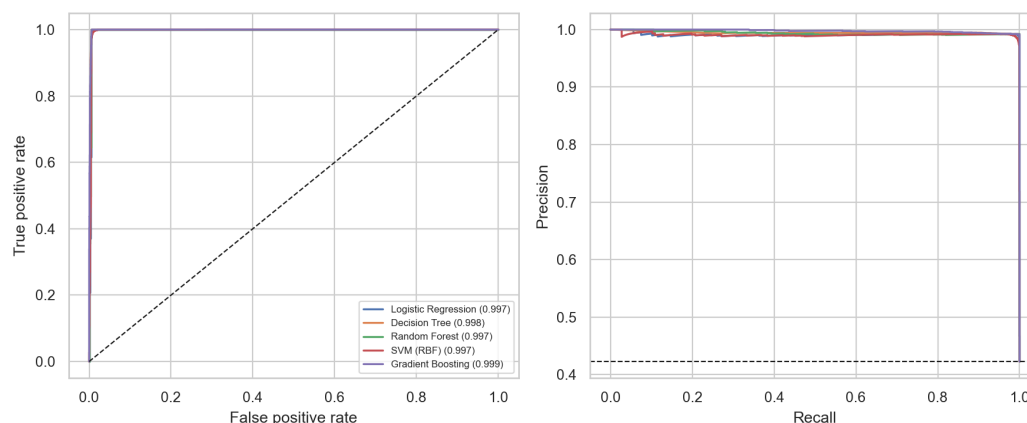
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What I did this week

This week I evaluated all tuned models once on the 6,723 test rows. Decision Tree and Random Forest are tied at F1 0.996 with recall 1.000. SVM is the weakest at F1 0.989 and also the slowest to train.

To check whether the differences are real I used McNemar's test on the test predictions. Decision Tree is significantly better than Logistic Regression and SVM (p below 0.001), but not different from Random Forest or Gradient Boosting.

Model	Precision	Recall	F1	ROC-AUC
Decision Tree	0.992	1.000	0.996	0.998
Random Forest	0.992	1.000	0.996	0.997
Gradient Boosting	0.992	1.000	0.996	0.999
Rule baseline	0.990	1.000	0.995	n/a
Logistic Regression	0.983	0.999	0.991	0.997
SVM (RBF)	0.986	0.993	0.989	0.997



ROC and precision-recall curves on the test set.

Next week

Test the effect of class imbalance handling, the decision threshold and each feature group.